IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 23 June 2025, accepted 14 July 2025, date of publication 7 August 2025, date of current version 15 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3596720

RESEARCH ARTICLE

Multimodality-Based Situational Knowledge for Obstacle Detection and Alert Generation to Enhance the Navigation Assistive Systems

RANIYAH WAZIRALI[1,3] AND EMAN ALKHAMMASH[2,3]
[1]College of Computing and Informatics, Saudi Electronic University, Riyadh 11673, Saudi Arabia
[2]Department of Computer Science, College of Computers and Information Technology, Taif University, Taif, Saudi Arabia
[3]King Salman Center for Disability Research, Riyadh 11614, Saudi Arabia

Corresponding author: Raniyah Wazirali (r.wazirali@seu.edu.sa)

This work was supported by the King Salman Center for Disability Research through the Research Group under Grant KSRG-2024-407.

ABSTRACT Recently, Deep Learning (DL) architectures have increasingly contributed to technological advancements, particularly for people with disabilities, by improving assistive systems. By adopting advanced DL models, intelligent assistive systems have recently become powerful with object detection, image captioning, and speech generation. The navigation system requires a concise and comprehensive alert generation that depends on an efficient caption generation model to describe the obstacles in the surroundings. This work introduces a Multimodality-based Situational Knowledge-aware (MuSK) system designed for obstacle alert generation in outdoor sidewalk environments by integrating visual and non-visual cues through contextual multimodal processing. To enhance obstacle detection, a layered attention mechanism fuses multi-sensory data with the visual feature maps extracted from the multimodal images, enabling obstacle-specific knowledge learning from diverse modalities. A critical innovation of this work is the personalized attention score, computed by a custom logic gate-based approach that dynamically adjusts obstacle weights based on situational knowledge based on multi-sensory data. Consequently, the logic gate mechanism refines attention scores by filtering and prioritizing relevant obstacle cues, improving obstacle detection accuracy. Furthermore, the cross-attention modeling of multimodal cues generates real-time, context-aware captions of detected obstacles, enabling precise situational awareness and safer navigation for the visually impaired. Thus, the experimental outcome reveals that the proposed MuSK system proves an accurate obstacle alert generation while tested on benchmark datasets, emphasizing the importance of personalized attention mechanisms and multi-sensory integration in improving obstacle detection in dynamic outdoor environments.

INDEX TERMS Visually impaired, image captioning, multimodal, multi-sensory, obstacle detection, obstacle-aware attention, obstacle alert generation, personalized attention, situational knowledge, YOLOv7, voice alert.

I. INTRODUCTION

The lifestyle and independence of Visually Impaired People (VIPs) have become the most important social concern. The advancement of technologies dramatically contributes to developing assistive systems. According to statistics from the World Health Organization (WHO), 285 million individuals have visual impairment and significant vision loss over the age of 50 [1]. Assistive technologies are crucial for the daily activities of the visually impaired, and intelligent technologies significantly improve the quality of their functional activities [2], [3]. Notably, electronic travel aids, such as smartphone applications or wearable devices, rely on mapping technologies to provide real-time auditory instructions and route guidance. Computer vision enables the system to perceive, understand, and interpret visually captured data

The associate editor coordinating the review of this manuscript and approving it for publication was Wonhee Kim.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

in which object detection [4] predominantly identifies and localizes specific objects within images captured by the visually impaired. In computer vision, image captioning is paramount in enhancing human-computer interaction and creating textual descriptions for visual images. Despite advancements, accurately describing complex scenes for the visually impaired remains challenging without sacrificing the computational efficiency of smart navigation wearables. In contrast to traditional image captioning architectures [5] that focus on image features, capturing object features enhances precision in real-world scenarios. Integrating object detection with image captioning significantly improves assistive systems by providing contextual object understanding. Even though deep neural networks have developed advancements in the object detection models [6], [7], [8], such as Single Shot Detector (SSD), Deconvolution SSD, M2Det, and RetinaNet, they struggle with detecting obstacles and small objects due to limited features and datasets. MobileNet is another object detection model designed by Google based on a family of Convolutional Neural Network (CNN) architectures to support computer vision tasks. Among them, You Only Look Once (YOLO) [9] offers real-time detection but still faces difficulties in dense outdoor environments due to the need for large-scale pretrained knowledge.

Traditional obstacle detection aids [10] rely on sensor-based technologies or computer vision algorithms to detect environmental obstacles. Assisting the visually impaired in real-world navigation requires diversified features in addition to image features, which demands situational awareness to understand obstacles within a specific time and spatial context. Achieving comprehensive situational knowledge requires multimodal data, combining images with sensor input. Previous works [11] have utilized multimodal cues to enhance navigation accuracy for the visually impaired. However, inaccurate obstacle detection is likely to compromise navigation assistance quality. Hence, balancing detection performance with a lightweight design is challenging due to the complexity of numerous objects, requiring large-scale pretrained models and generating captions for all recognized objects. To overcome these constraints in the encoder-decoder image captioning architecture, this work utilizes multimodal inputs, including visual and non-visual cues, and designs a novel encoder-decoder architecture for the obstacle alert system. The proposed obstacle alert system has three components: a feature extractor component with custom object feature maps extraction, an encoder with situational knowledge-aware obstacle knowledge learning through layered attention, and a decoder with caption generation for obstacles.

The notable contributions of the Multimodality-based Situational Knowledge-aware obstacle alert (MuSK) system are as follows.

- The proposed obstacle alert system integrates situational knowledge from visual and non-visual cues through contextual multimodal processing to enhance obstacle detection in outdoor sidewalk environments.
- The proposed approach is designed for the improved YOLOv7 model by integrating MobileNet for efficient object feature extraction from multimodal images, particularly infrared and depth images, for accurate and robust obstacle detection in complex environments.
- Notably, the system is integrated with multi-sensory data, including Light Detection And Ranging (LiDAR) and digital compass, providing a comprehensive understanding of the environment surrounded by the visually impaired for obstacle region detection, designed in a layered attention model that fuses multi-sensory data with visual feature maps.
- Moreover, a novel custom logic gate-based approach assisted the computation of personalized attention scores from the multi-sensory situational knowledge, dynamically alerting the obstacle regions in a real-time environment by prioritizing the neighborhood obstacles to the visually impaired.
- Finally, the cross-attention modeling of multimodal cues in the text generation model produces obstacle-aware captions, improving situational awareness and navigation safety in the outdoor environment.

II. LITERATURE REVIEW

This section presents an in-depth review of object detection, obstacle detection, and multimodal object detection systems.

A. OBJECT DETECTION APPROACHES

Deep Learning (DL) architectures have revolutionized object detection tasks by applying their ability to detect intricate patterns in a complex real-time environment. The work in [12] presents the DETR with an Improved deNoising anchOr box (DINO) model, which is the end-to-end object detector based on the improved denoising anchor in the DEtection Transformer (DETR). The Self-Learning Multi-Scale object detection (SLMS) approach [13] extends the SSD model, balancing spatial and semantic information by enhancing shallow features to improve the modeling of tiny objects with informative patterns. Moreover, self-learning-based feature selection and feature fusion methods assist in determining the importance of features for accurate object detection. The research work [14] employs the YOLOv5 model to detect vehicles under different weather conditions, which include cars, traffic lights, and pedestrians.

The brief study of the Site Object Detection dataset (SODA) is presented in work [15], gathered from various construction sites, incorporating multiple object classes under the layout, machine, material, and worker. By utilizing the SODA, object detection systems in the construction site have increasingly gained benefits regarding accurate object recognition. The YOLO object detection algorithm [16] detects prostate cancer by localizing and classifying the tissue patterns in the medical field. By re-training the YOLO algorithm with the prostate biopsy tissue patterns and finetuning the hyperparameters, it effectively detects the objects in the medical images for diagnosis. The

research in [17] and [18] integrates object detection with image captioning for the input images. The image-adaptive YOLO (IA-YOLO) model in [19] enhances the images to detect hazardous objects even in hostile weather conditions, such as foggy environments. The work in [20] designs a YOLO model integrating Adaptive Frame Control (AFC). Subsequently, the object detection model supports efficient real-time processing in the YOLO model mounted with the network cameras; however, detecting tiny airborne objects is still challenging.

To resolve this constraint, a research paper [21] presents a Tiny Airborne object Detection (TAD) that employs a heuristic motion description method to effectively achieve fast pixel-level motion to detect tiny airborne objects with limited appearance information. To quickly and safely navigate the VIP in indoor and outdoor environments, the work in [22] designs a lightweight CNN model for object recognition. The work in [23] investigates the objects in the perspective of the motion features over time in the image and ignores the noisy motion features to learn the object patterns by the CNN model, which leverages the accurate generation of the image captions. The researchers [24], [25], [26] employed the different versions of the YOLO model to extract the features and classify the objects to assist the VIP. The custom object detection model in [27] combines the advantages of transfer learning and a pre-trained model to improve the effectiveness of object detection. Initially, it employs pretrained models, such as the ResNet or VGGNet by Visual Geometry Group (VGG), to extract the high-level abstractive features. Then, it finetunes the custom target task with the extracted features.

B. OBSTACLE DETECTION APPROACHES

An outdoor navigation system in [28] examines the data from wearable devices and provides services on low-cost devices. Summarizing a three-dimensional perspective of the obstacle position enhances the perception of the VIP with the aid of the DL-based 2D object detector. The multi-sensory guidance system in [29] employs the Oriented Fast and Rotated Brief Simultaneous Localization and Mapping (ORB-SLAM) and YOLO models to generate the navigation map and provide path planning against the obstacles. The work in [30] presents an assistance algorithm by integrating object recognition and obstacle detection using the YOLOv3 model. The work in [31] presents an outdoor navigation solution for the VIP using the smartphone camera and DL model.

By detecting the objects and obstacles in the captured images based on a set of matching pixels and adjusting the same color, it enforces voice guidance for autonomous walking in the outdoor environment. The work in [32] relies on intelligent walking sticks and wearable smart glasses to guide VIPs with an intelligent assistive system with fall detection and obstacle avoidance. Moreover, mobile applications and cloud-based data management platforms leverage the avoidance of aerial obstacle collision accidents by triggering vibration on a walking stick.

Smart_Eye model [33] combines artificial intelligence-based smart applications and sensor technology, enhancing the VIP quality through ultrasonic sensorbased obstacle detection and voice command-based navigation. An intelligent navigation assistance system in [34] exploits the Neural Architecture Search (NAS) and DL. The work in [35] assists visually impaired people in tracking obstacles in their navigation path by designing a Smart-alert Walker and Smart-fold Cane for their smart shoes and stick, respectively. A lightweight shoe, the Smart-Alert Walker serves as an emergency device, alerting its wearer to obstacles within a brief range. In contrast, the stick applies the SSD mechanism for object identification and the Recurrent Neural Network (RNN) model for classification and notification generation.

C. MULTIMODAL OBJECT DETECTION APPROACHES

The work in [36] applied the YOLOv5 model to detect objects with a third-eye design for the VIP. Training the YOLO model with a large-scale image captioning dataset precisely generates the captions and audio alerts for safer navigation. Multimodal DL architecture in [37] presents two stages, objectness estimation and region-wise object recognition, which jointly detect the Red, Green, and Blue (RGB) channels with depth objects by examining the RGB and depth images with the correlative relationships. The weighted mean-based adaptive object detection approach [38] enhances detection performance by combining the object detection outcome from the RGB camera and LiDAR-based images in autonomous driving. Applying the YOLO model detects the objects from the image data and Point Cloud Data (PCD) and fuses the detection outcome based on the weighted scheme.

W3Net model [39] partitions the pedestrian detection task into the Where, What, and Whether problems for pedestrian localization, scale prediction, and classification, respectively. Multi-label Convolutional Support Vector Machine (CSVM)-based scene description approach [40] models a set of linear SVM algorithms as the filter banks to generate the feature map. The smart navigation approach [41] initially augments the images and annotates the images with objects to improve the robustness of the DL model. In the subsequence of object recognition, the obstacle detection model generates the auditory data as the alerts during navigation. The image captioning model in [42] employs a Neural Architecture Search Network (NASNet) to represent the image patterns and employs the Radial Basis Function Neural Network (RBFNN) to describe the images. Moreover, applying the Red Deer Optimization (RDO) and Butterfly Optimization Algorithm (BOA) optimizes the NASNet and RBFNN model parameters for quick guidance.

III. SYSTEM MODEL AND PROBLEM FORMULATION

In the Outdoor environment, navigating the visually impaired
In the Outdoor environment, navigating the visually impaired
poses unique challenges in the perspective of safe navigation

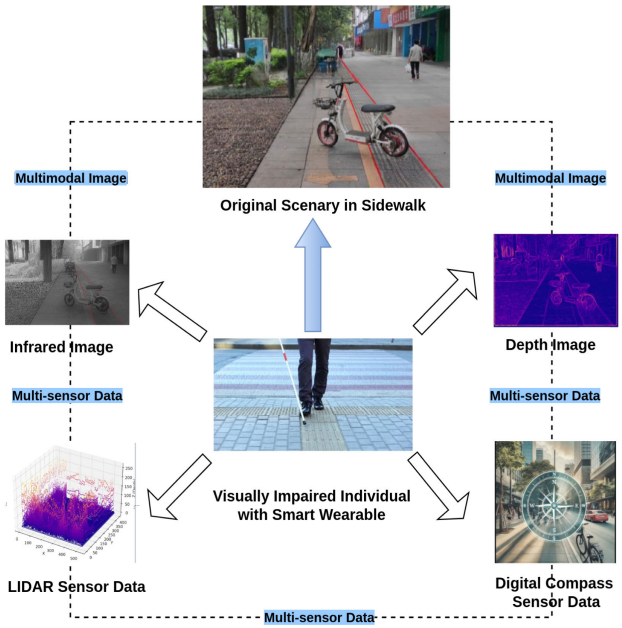

FIGURE 1. Multimodal and multi-sensory data in obstacle alert system.

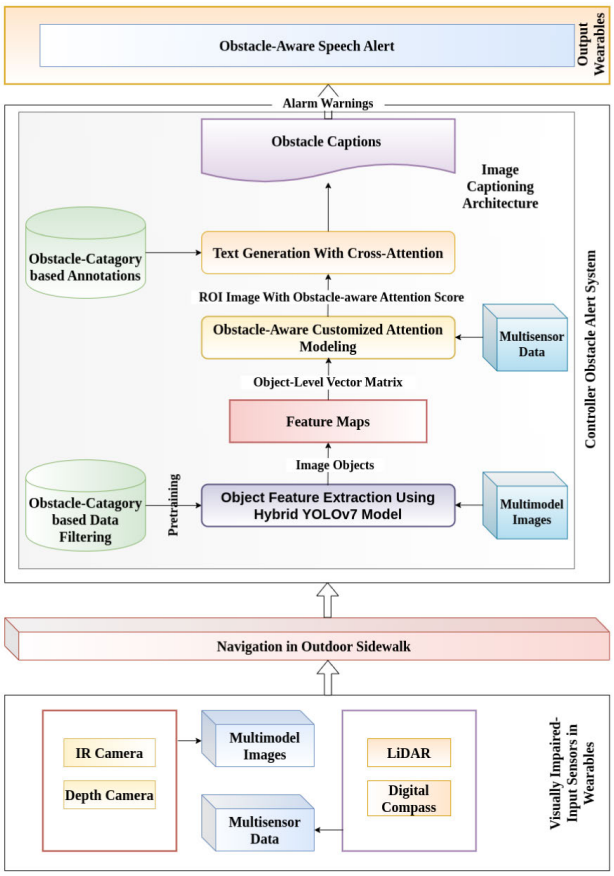

FIGURE 2. System design of the proposed obstacle alert system.

due to obstacles in their path. In the real world, ensuring safe navigation for the visually impaired requires real-time obstacle recognition. Advances in computer vision allow object feature extraction models to identify obstacles from camera images on navigation devices. Beyond object recognition, this work enhances human-visual interaction by delivering obstacle-related information. Safe navigation is achieved through path planning that integrates environmental conditions, sensory data, user preferences, and obstacle locations. In the real world, outdoor navigation assistive systems have been increasingly applied for journey planning, environment mapping, and real-time navigation [43]. This work designs an obstacle alert system for outdoor navigation using computer vision to assist the visually impaired. The system adapts the navigation strategies based on real-time obstacle detection, providing visually impaired users with a situational understanding of their surroundings. This work designs the obstacle detection system for the Pathways or Sidewalks outdoors, including the person, stop sign, animals, fire hydrant, stairs, and bench as the obstacle objects in the Microsoft Common Objects in Context (MS-COCO) dataset. The proposed obstacle detection system assists the visually impaired with the obstacle alert when the individuals are navigating within the boundary of the sidewalk with the assistance of multimodal images and multi-sensory data, as depicted in Figure 1. In the context of outdoor O_{OD}, sidewalk incorporates various elements, such as the person, stop sign, animals, fire hydrant, stairs, and bench. Hence, the outdoor elements are represented as, $O_{\mathrm{OD}} = \{E_i^{\mathrm{OD}}, \ldots, E_m^{\mathrm{OD}}\}$. For the incorporation of situational knowledge in the obstacle alert system, this work utilizes multi-sensory inputs in addition to the images obtained from the RGB camera and Infrared (IR) through the smart wearables, such as a smartwatch, eyeglasses, and headset of the visually impaired. In particular, the non-vision sensors, such as the digital compass and LiDAR, strengthen the decision-making of obstacle detection and facilitate concise caption generation. Finally, the assistive system relies on smartwatches and headsets to assist the VIP via speech warning. The developed obstacle detection module has to be deployed on the intelligent wearables associated with the navigation system, ensuring lightweight, affordable, and easy-to-access services for the visually impaired. Figure 2 illustrates an outline of the proposed obstacle alert architecture.

This work addresses several constraints to effectively detect the obstacles and generate alerts for the visually impaired during navigation in an outdoor sidewalk environment.

A. PROBLEM FORMULATION

Identifying and avoiding environmental obstacles is essential for blind navigation systems to ensure safe transit. Defining obstacles from visual objects in outdoor paths is challenging due to varying types, sizes, and locations. Understanding obstacles is difficult without visual cues, making image captioning a promising tool for improving navigation through obstacle detection. However, providing VIPs with rich and

descriptive environment information remains challenging. Traditional object feature extraction methods pretrained on large-scale datasets impact training efficiency due to irrelevant categories. Moreover, custom feature extraction is necessary to enhance caption generation for smart assistance systems. As a result, the outdoor obstacle detection model necessitates examining situational knowledge from the different sources surrounding the visually impaired.

1) GENERALIZED MULTIMODAL OBJECT FEATURE EXTRACTION

Object feature extraction with minimized localization loss is critical for the detected objects, as modeled in equation (1). In equation (1), IoU is the Intersection over Union between the predicted box (P) and target box (T).$lP \wedge lT$ and CT_brefers to the central points of the P and T boxes, respectively.lP_lT denotes the localization vectors of the predicted and target boxes with the central coordinates of 'x' and 'y', respectively. $\mathrm{DL}_{P\text{-}T}$ is a diagonal length of the smallest box that covers the predicted and target boxes in a particular image 'i'.

$$\min_i \left[[1 - \mathrm{IoU}] + \frac{\alpha^2 \left[\mathrm{CP}_b \left(lP^{C_x}, lP^{C_y} \right), \ \mathrm{CT}_b \left(lT^{C_x}, lT^{C_y} \right) \right]}{(\mathrm{DL}_{P\text{-}T})^2} \right] \tag{1}$$

Even though conventional SSD and YOLO models recognize the objects' feature maps in each image (i), determining the class-specific object is paramount for obstacle detection with minimal false positives.

$$O_i = \{RO_1, \ RO_2, \ \ldots, \ RO_n\}^C \tag{2}$$

Hence, to augment the perception of the visually impaired, exploring or utilizing the multimodal images and multiple senses with custom object feature extraction is essential to adapt to the dynamic outdoor environment.

2) VAST IMAGE CAPTIONING

Generating the captions for all the recognized objects' features misguides the visually impaired with the massive alert generation, which is likely to ignore the essential warnings. To overcome this constraint, objects are to be selected from all the recognized objects for each particular image, which intends to focus on object filtering.

$$O_k = \{RO_1, \ RO_2, \ \ldots, \ RO_n\} \cap \{Ob_1, \ Ob_2, \ \ldots, \ Ob_m\} \tag{3}$$

$$O_k \subseteq O_i \tag{4}$$

As formulated in equations (3) and (4), refers to the filtered objects that are highly relevant to detecting obstacles for the navigation systems.

3) LIGHTWEIGHT OBSTACLE DETECTION

Obstacle detection is a significant constraint for the YOLO model without the comprehensive dataset about the obstacle detection task. While combining object extraction and image captioning aids visually impaired navigation, it increases the computational requirements, complicating the lightweight design. Deploying YOLO on wearable devices is difficult due to its size and processing needs, and pretrained models for image captioning also require substantial resources. Timely alerts are essential for visually impaired users, making lightweight obstacle detection crucial for achieving real-time performance on resource-constrained devices.

B. PRELIMINARIES

The MuSK system employs image captioning encoder-decoder architecture with a personalized obstacle-aware attention score to enhance navigation assistance through the obstacle alert system, shown in Figure 3. It initially employs the YOLOv7 [44] to design this to detect objects from the captured outdoor sidewalk images. YOLOv7 is a single-stage, highly accurate, and fastest real-time object detector, a Fully Connected Neural Network (FCNN) based convolutional architecture.

1) AN IMPROVED YOLOV7

For the proposed obstacle alert generation system, to design a lightweight object feature extraction model for the captured images by the visually impaired, the YOLOv7 model is enhanced with the MobileNetv3 model as the backbone feature extraction network. Since the backbone of the YOLOv7 model affects the parameters and decides the speed of the detection model, a compact YOLOv7 model is designed. Thus, the proposed approach replaces the traditional convolution operation with the deformable convolution [45] that adaptively changes the shape of the convolution kernel for the input image. Owing to the predefined shape and size of the kernel in the traditional CNNs, the adoption of deformable convolution facilitates the adaptability of the network for the input image with different shapes and spatial positions, ensuring the reduced number of parameters and kernels without compromising the performance of convolution operations.

For the training of the YOLOv7 model, the captured input image is set as 640 × 640 size after preprocessing the raw image in terms of noise filtering, cropping, and sharpening. The proposed approach designs a lightweight YOLOv7 model with the MobileNetv3 backbone network and deformable convolution without compromising detection accuracy to reduce the storage space during object feature extraction in outdoor locations. Consequently, an improved model ensures fewer parameters in the YOLOv7 model for an individual's resource-constrained hardware devices. In the proposed lightweight YOLOv7 model, MobilenetV3 exploits the h-swish activation function to mitigate the number of operations and improve the model performance. Moreover, the deformable convolution network randomly initializes the 1 × 1 convolution to reduce the channel dimensions, transforming the 32 pixels to 16 pixels.

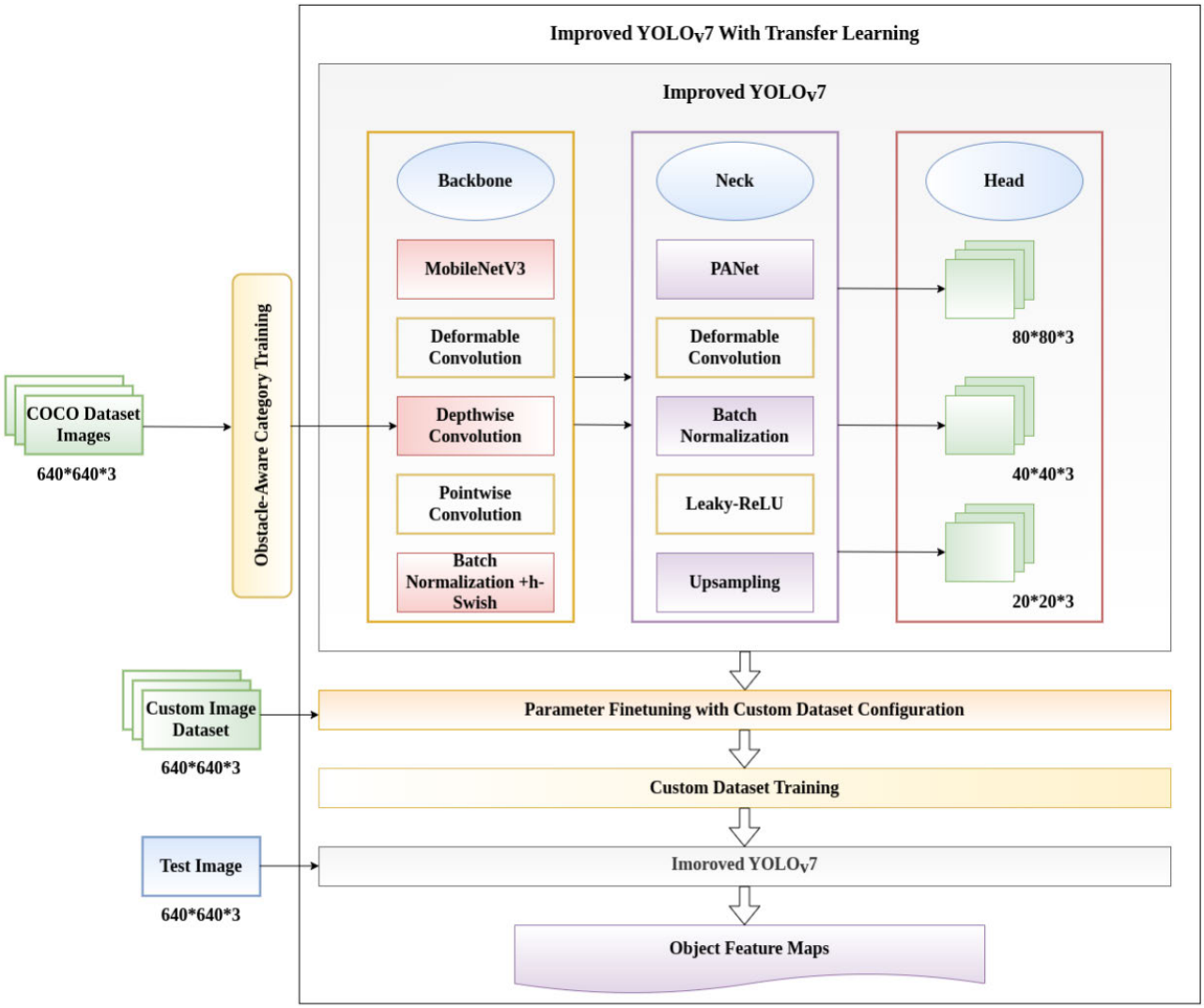

FIGURE 3. A lightweight YOLOv7 model training with transfer learning.

Moreover, to train the custom navigation region dataset with the YOLO knowledge, the proposed approach designs the improved YOLOv7 with the transfer learning that provides the initial training knowledge for immediate and accurate object feature extraction. In addition, transfer learning paves the way for improving the model performance without requiring a massive amount of image frames by utilizing the pretrained weights from the large-scale image dataset, such as the MS-COCO dataset. In the improved YOLOv7 model, the learned MS-COCO features are finetuned for the target dataset with updated parameters, resulting in minimal computation during training and testing of the target images. Thus, the design of backbone and convolutional layers for feature extraction and transfer learning ensures lightweight processing.

IV. PROPOSED MuSK SYSTEM: SITUATIONAL KNOWLEDGE-AWARE OBSTACLE DETECTION

The proposed system personalizes the attention layer to contextually learn and encode object features as obstacle weights, enhancing caption generation. The proposed design includes hybrid object feature extraction, multi-sensory obstacle-aware cross-attention, and ROI-based image captioning. Visually impaired users utilize smart glasses, watches, and headsets, integrating situational knowledge from sensors for obstacle detection. Multimodality, combining visual and non-visual cues, such as images and sensory data, improves situational awareness. Thus, the multimodal system detects obstacles using image-based object extraction and sensory data filtering, providing real-time alerts during outdoor navigation. As depicted in Figure 4, MuSK enables the

visually impaired to understand the environment by utilizing the situational knowledge of multimodal cues.

The operational flow of the proposed approach is described as follows. The proposed approach works as follows: A visually impaired individual, 'Person A,' uses smart wearables equipped with RGB and IR cameras, LiDAR, and a digital compass while navigating outdoors. The MuSK system captures RGB and depth images, fusing visual cues through element-wise multiplication to extract object features using two improved YOLOv7 models. Multimodal fusion integrates RGB and depth data to recognize ground-level objects, such as a person, stop sign, animal, and stairs. LiDAR and compass inputs are then used to highlight obstacles by calculating distances from 'Person A'. For example, objects within 1 meter, like people or animals, are tagged as obstacles. A layered attention model computes an obstacle-aware weight matrix, linking multi-sensory inputs with personalized attention scores. Finally, the captioning model generates descriptions of these obstacles using the learned situational knowledge from visual and non-visual cues.

Figure 5 shows the demonstrative process of the proposed obstacle alert system. In the proposed system, the obstacle alert generation is performed by image captioning that is enhanced by YOLO-based object feature extraction and obstacle-aware attention design, which are briefly described below.

A. FEATURE EXTRACTOR: MULTIMODAL IMAGE-BASED OBJECT FEATURE EXTRACTION

For the design of the image captioning encoder, the proposed multimodal system utilizes the multimodal images as the visual cues for the object feature extraction, particularly from the RGB and depth images in which images are captured by the IR and depth camera integrated into the smart wearables.

The object feature extraction leverages infrared and depth images to capture sharper features with minimal noise. Multimodal extraction improves robustness and accuracy in challenging scenarios like occlusions, lighting variations, and clutter.

The system incorporates multimodal images and enhances obstacle alerts with richer contextual information. To model feature distributions, IR and depth images are independently processed using an improved YOLOv7 model, applying transfer learning to train and test the navigation dataset. Subsequently, both the learned representations are combined by a late fusion method in the task of contextual multimodal fusion, as depicted in Figure 6.

Initially, the IR and depth images are scaled into a particular image size that is fitted for the learning model. The original IR and depth images are resized using consistent scaling operations like image warping. Assuming joint and individual representations in the transformed feature space, multimodal feature learning analyzes feature maps from two improved YOLOv7 models. Instead of early fusion, intermediate feature representations capture contextual information about object textures and shapes. The multimodal correlation utilizes feature maps from the end of the (k+1)th layer in the YOLOv7 network. The final convolutional layer then adjusts the depth of each feature map, enhancing modularity and flexibility between the two YOLO networks. To generate the correlated feature map ($h\left(Z_{ij}\right)$) of multimodality, the element-wise multiplication is applied for the feature maps of RGB and depth modalities using the Hadamard product.

$$h\left(Z_{ij}\right) = h\left(X_{ij}\right) \circ h\left(Y_{ij}\right) \tag{5}$$

In equation (5), the hidden feature maps of the RGB images and depth modalities (M_1, M_2) are, $X_{ij} = \left\{1, \ldots, M_1^{FM}\right\}$ and $Y_{ij} = \left\{1, \ldots, M_2^{FM}\right\}$, respectively. Consequently, an improved YOLOv7-based multimodal object feature extraction model jointly learns the object feature maps from the consistent representation of RGB and depth modalities, which enhances the quality of obstacle detection. The outcome of the feature extractor in the image captioning architecture is defined by the improved YOLO model, presented in equation (6).

$$b_i = \{x, y, W, H, C_{\text{score}}, P(C)\} \tag{6}$$

For each image's grid, the feature vector (b_i) returned by the hybrid YOLO model, which includes the bounding boxes and object class probability. The bounding box is the composition of grid cell coordinates (x,y) and width-height (W, H) of the bounding boxes, along with the confidence score. Thus, in the MuSK system, the design of an improved YOLOv7 model extracts the object feature maps, which are further provided to the attention layer for obstacle detection within the image captioning encoder-decoder architecture.

B. ENCODER: MULTIMODAL AND MULTI-SENSORY DATA-BASED SITUATIONAL KNOWLEDGE LEARNING

In image captioning, attention plays a significant role, defining the learning focus of the model within the image when generating the textual descriptions. It utilizes feature vectors from visual and non-visual cues to generate captions for filtered obstacles. This approach learns the multimodal cues through layered attention, designing personalized attention based on obstacle knowledge from surrounding sensor inputs. Moreover, the proposed system personalizes the attention scores using a custom obstacle-weight matrix based on multi-sensory inputs. In particular, to improve the perception ability and accuracy of obstacle recognition in different contexts, the proposed obstacle modeling system utilizes multi-sensory data for the attention design of the image captioning model [46]. The multi-sensory data involves the LiDAR sensor and digital compass information in the proposed obstacle-aware caption generation system. Figure 7 illustrates obstacle-aware attention in encoder-decoder captioning architecture. Initially, the proposed approach examines the location of an individual and captures a set of object feature maps statically detected by the improved YOLOv7 model. In the dynamic context, obstacle detection for caption

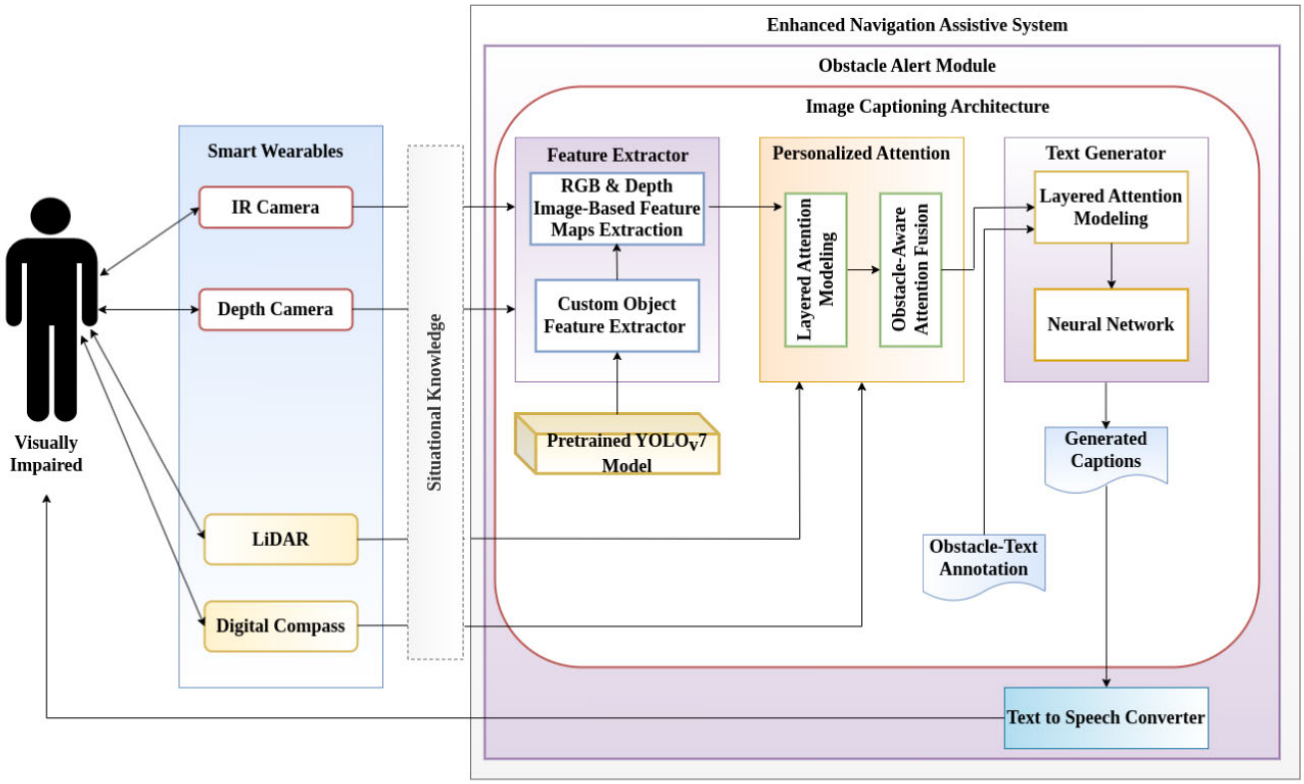

FIGURE 4. Modeling situational knowledge in MuSK obstacle alert generation.

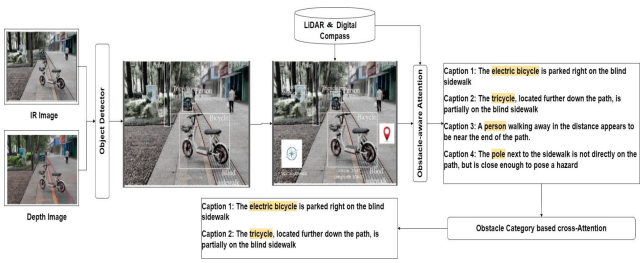

FIGURE 5. Illustration of obstacle alert system.

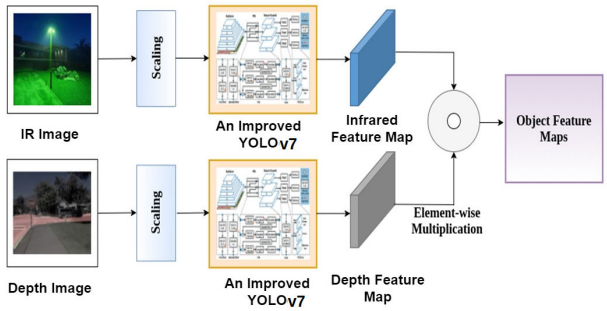

FIGURE 6. Visual cues-based multimodal object feature extraction.

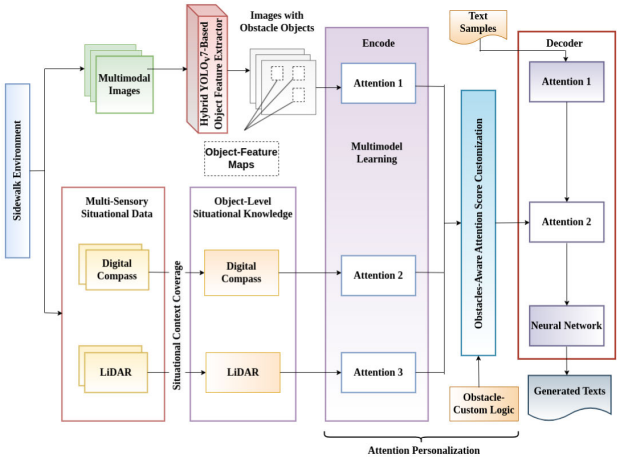

FIGURE 7. Obstacle-aware attention in encoder-decoder captioning architecture.

generation relies on the distance between the camera and objects across sequential images, determined using LiDAR and encoded in the obstacle-aware weight matrix. The MuSK system uses data from a Time-of-Flight (TOF) LiDAR sensor with a 4-meter range to identify objects as obstacles when they are within 1 meter. The LiDAR, covering 240° around the user, is mounted on headbands, canes, and boots to detect obstacles in the navigation path. Closer objects in time-varying frames are prioritized as obstacles, with multi-sensory inputs integrated via mid-level fusion and attention scores based on temporal-spatial relationships. In contrast, the multi-sensory inputs are integrated by the early fusion at the data level for the obstacle-aware weight

matrix modeling, referred to as the bias matrix.

$$A_{b(i)}(t) = f_A\,(h_t,\ b_i) \tag{7}$$

Equation (7) formulates the attention score of the fth feature vector (b_i) in the ith image at timestep 't' for the object feature maps. f_A is a function of the relationship between the hidden vector (h_t) and a subset of the image feature vector (b_i). For real-time processing, the logic gate provides high-speed performance with deterministic outputs without consuming enormous resources, as mentioned in Table 1, in which '0' denotes the non-obstacles and '1' represents the obstacles.

To further enable the Region-Of-Interest (ROI) processing, this approach attempts to detect the obstacles from the massively detected objects. Object filtering for obstacle detection relies on key factors such as height threshold, LiDAR data, and spatial-temporal context. The obstacle-aware weight model processes LiDAR and digital compass data within the navigation area at specific times. Spatial filtering is achieved through segmentation and temporal filtering via windowing to refine object feature maps. Logic gate-based obstacle weights in the custom attention score model highlight obstructive objects while fading irrelevant ones, optimizing obstacle detection for caption generation in the encoder-decoder architecture. As formulated in equation (8), to compute the ROI region of each image, the proposed approach computes the obstacle weight ($\mathrm{Obs}_w^{i(s,t)}$)and confidence rate (C_R) of each object's feature map (b_i).

$$\mathrm{ROI}_i^b = \mathrm{Obs}_w^{i(s,t)} \times A_S\,(b_i) \tag{8}$$

In this image captioning architecture, the attention mechanism enables the decoder part of the image captioning to focus precisely on the obstacle regions by combining the attention score. $A_S(b_i)$ returned by the layered attention with the computed obstacle-aware weight matrix. In the task of multisensory-based obstacle weight modeling, this approach applies a Custom Logic Gate (CLG) that follows a unique combination of logical operations among multi-sensory factors ($F_1(t),\ F_2(t),\ F_3(t)$) at time 't', described in equation (9).

$$f_A \Rightarrow \mathrm{Obs}_w^{i(s,t)} = \mathrm{CLG}\,(F_1(t),\ F_2(t),\ F_3(t)) \tag{9}$$

From the results of the spatial and temporal coordinates (s,t) mapping for the object feature extraction outcome, the logic gate produces the obstacle weight that is tagged to each object in which a higher weight represents that the object is an obstacle in a particular image frame after normalizing the attention weights with the softmax.

As mentioned in Table 1, the person-object height-width threshold is based on the horizontal and vertical coordinates from the ground-level surface to the detected objects, in which the threshold is based on the person's height. If an object's height or width exceeds the threshold, its logical value is set to '1'; otherwise, it is '0'. Similarly, LiDAR intensity values above the threshold result in '1', while lower values are '0'. When spatial and temporal sensor inputs match object coordinates, and the distance is minimal, the value is '1'; otherwise, '0'. Obstacle weights are then computed for each object's feature map based on these logical factors, customizing attention scores in the layered attention mechanism of the encoder-decoder captioning model. Consequently, the obstacle-aware attention weight ($A'_{b(i)}(t)$) in the image captioning architecture is computed before modeling the context vector ($C_t(i)$) of the image frames as the input to the decoder model, the caption generator.

TABLE 1. Computation of Obstacle-aware bias matrix for personalized attention score modeling.

Person-Object Height-Width Threshold	LiDAR Intensity	Digital Compass - Timestamp	Obstacle Weight
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

$$A'_{b(i)}(t) = A_{b(i)}(t) \cdot \mathrm{ROI}_i^b \tag{10}$$

$$C_t(i) = \sum_{b=1}^{B} A'_{b(i)}(t) \cdot b_i \tag{11}$$

Accordingly, equation (11) computes the context vector based on the summation of attention weights for the objects referring to the feature maps (b_i) in which 'B' denotes the number of recognized object-level feature maps in the image frame. From the inspiration of the work [47], this approach recognizes the obstacles in the outdoors based on the sequentially captured images. The proposed approach predominantly focuses on generating obstacle alerts during navigation rather than utilizing all the objects for efficient alert generation through lightweight caption generation. In the image frames, the obstacle-recognized image regions are contemplated as the ROI for the obstacle alert generation. Thus, customized attention modeling indicates that highlighting obstacle regions by the layered attention of the decoder in the image captioning enforces caption generation only for the obstacles.

C. DECODER: ROI-BASED IMAGE CAPTIONING WITH CROSS-ATTENTION

For situational knowledge-based navigation assistance, the proposed caption generation model utilizes the outcome of the improved YOLOv7 model by highlighting ROI-based image features modeled by the obstacle-aware attention score customization.

In the MuSK system, the caption generation model is a composition of an improved YOLOv7 feature extractor for the layered attention-associated encoder and the sequential neural network as the decoder, incorporating obstacle-aware cross-attention modeling across the multimodal cues. By pre-training the encoder-decoder model on the obstacle-category filtered MS-COCO knowledge that comprises the images

Algorithm 1 Pseudocode of the Proposed Obstacle Alert Generation

```
Input  : Visually Impaired Individual with Multimodal Cues
Output: Obstacle alert
for all captured multimodal (RGB+Depth) images (i) do
    Apply image preprocessing
    // An Improved YOLOv7 //
    MobileNetV3 and Deformable Convolution-based Pattern Learning with transfer learning
    // Feature Extractor (Visual Cue): Mulitmodal Object Feature Extraction //
    Extract visual-level Features for RGB and Depth images using two YOLOv7 with custom pretraining
    for all extracted features of i^th image (RGB+Depth) do
        Apply the element-wise multiplication ;
        Detect object feature maps;
        // Attention: Multi-sensory based Situational Knowledge learning //
        Capture LiDAR sensor, Digital compass, and GPS-sensed data dynamically
        for all the object-level feature maps do
            Design layered attention for multimodal cues using equations (8)-(11)
            Compute obstacle weight as bias matrix using CLG and personalize attention with bias matrix
        end
        Detect the ROI in each image frame
    end
    // Decoder: ROI and COCO-assisted Caption Extraction //
    for the obstacle-aware sample categories in COCO dataset do
        Pretrain the text generation model with custom text-level training knowledge;
    end
    for the pretrained text generation model do
        for all the ROI-based filtered images do
            Execute the decoder model with cross-attention of multimodal cues using equations (12)-(13)
            Generate the caption tags
        end
    end
    Convert generated captions into speech as the obstacle alert;
end
```

with the captions, the decoder model utilizes the tags for the filtered object patches from the text-level image features with the dimension of $TF \in F_{N\times K}$ and generates the captions for the ROI image based on the outdoor features learned from the raw visual-level features with the dimension of $VI \in F_{N\times M}$ through cross-attention modeling. Where 'N', 'M', and 'K' denotes the number of images at a particular time 't', the number of object-level text samples in each image (i), and several obstacle-aware weight-associated attention scores, that is $K \subseteq M$ in each image (i) respectively. In the image captioning architecture, the decoder component comprises two attention layers for text-level features and the encoded multimodal cues-based context vector in a layered sequential attention manner, referred to in equations (12) and (13).

$$A'_{TF_i}(t) = A_{TF_i}(t) \cdot \mathrm{ROI}_i^b \tag{12}$$

$$T_{\mathrm{Decoder}}(i) = A'_{TF_i}(t) \cup A\left[C_t(i)\right] \tag{13}$$

In equation (13), '$\cup'$ refers to the sequential consideration of the textual-level features, (TF) and Obstacle-aware context vector $(C_t(i))$ in the attention for the ith image, a decoder process during text generation. In the proposed caption generation model, the neural network model jointly learns the sequential representation of text-level features with the context vector as the mid-level fusion of feature fusion in the contextual multimodal fusion through cross-attention modeling. Thus, the proposed caption generation model learns the embeddings belonging to the object-level feature maps to generate a concise and comprehensive caption based on the improved YOLOv7 knowledge and the obstacle-aware attention weights. The algorithm flow of the proposed situational knowledge-aware obstacle alert model is presented as the pseudocode from steps (1 to 25) in Algorithm 1.

Finally, the image captions are converted into the voice alert sent to the wearables of smart headphones in the navigation assistance system.

D. DESIGN OF LIGHTWEIGHT OBSTACLE ALERT SYSTEM

In the proposed system, YOLOv7-assisted object feature extraction based on infrared and depth images improves the accuracy and efficiency of visually impaired people. In the multimodal images, IR and depth images are processed by YOLOv7, which is prominent for its speed and high object

detection accuracy in identifying and categorizing objects in real time. Combining these multimodal images lowers the computation burden and leverages the efficiency of YOLOv7, which is highly suitable for wearable devices and enables accurate obstacle detection in various illumination scenarios.

Moreover, LiDAR enhances the image frame configuration by providing accurate 3D mapping for obstacle identification, and a digital compass offers real-time geolocation and direction guidance. Consequently, combining multi-sensory data produces a lightweight, energy-efficient system that provides precise obstacle identification and navigational assistance without compromising the wearable devices' power consumption. In particular, by prioritizing different types of data from different sensors, the logic gate-controlled multimodal attention personalization helps to maintain the system operation efficiently.

With the adoption of a small YOLO model size, the convolution operations of O(n) complexity are based on the 'n' number of pixels in the input image in which the lightweight MobileNetV3 model has the O(n) complexity while reducing the depthwise separable convolution complexity.

During the obstacle object detection, LiDAR-based point cloud data has the complexity of O(m) for the 'm' number of points in the cloud of the image frame, whereas the digital compass has the complexity of O(1), which is negligible. Instead of $O\left((k \cdot d^2)\right)$ complexity in the traditional transformer encoder-decoder for 'k' input size and 'd' hidden dimension size, the proposed approach has O(k.d) complexity by the custom-logic-based attention mechanism. Thus, the proposed system proved its lightweight design for the smart wearables of the visually impaired individual.

V. RESULT ANALYSIS

In the context of performance evaluation, the experimental model validates the performance of object detection, obstacle detection, and image captioning. The experimental evaluation employs the MS-COCO dataset [48] as well as the Obstacle dataset [49] to exemplify the detection and caption generation performance of the generalized and contextual blind sidewalk dataset with the obstacles, respectively. The MS-COCO dataset comprises 164K indoor and outdoor images with annotations in the 80 object categories.

Moreover, to experiment with the multimodal inputs, the experimental model synthetically maps the LiDAR and digital compass values for a particular region of the outdoor environment, assumed as the data captured by the wearable devices. The blind sidewalk obstacle dataset [49] contains 7915 images with 40000 above bounding boxes in which the training set comprises 5066 images, the test set comprises 1583 images, and the validation set comprises 1266 images. The obstacle dataset consists of 15 obstacle types, and the categories belong to the sidewalk navigation of blind or visually impaired people. Table 2 provides the model parameters used during the implementation of the proposed system.

TABLE 2. Model parameters of the proposed system.

Parameter	YOLOv7 with MobileNetV3	Transformer
Learning Rate	0.0001	0.0001
Number of Heads	2	2
Activation	Leaky ReLU	ReLU, Sigmoid
Loss Function	CIoU loss, Binary_Crossentropy	Binary_Crossentropy
Epochs	100	100
Batch Size	32	64
Optimizer	Adam	Adam

Lightweight Parameter Design: To develop the lightweight design of the proposed obstacle alert system, this work focuses on several key implementation parameters. For the implementation of YOLOv7 with MobileNetV3 in the task of object feature extraction, 6 million and 2.5 million parameter sizes with the computation cost of 5.8 Giga Floating Point Operations per Second (GFLOPs) and 0.5 GFLOPs are modeled, respectively to support the real-time navigation through wearables. Moreover, the multi-sensor data, such as the LiDAR and digital compass, consumes less than 0.1 GFLOPs, ensuring minimal complexity.

The key innovation of the custom attention logic is based on a few custom rules, which mitigates the computation complexity in a complex neural network by basic thresholding and comparison among potential sensor recordings in the logic gates, speeding up real-time decision-making.

Finally, the proposed system is implemented with around 19 million parameters with computation load under 6GFLOPs and inference time ranges between 3 to 5 milliseconds, which enables efficient real-time processing on the smart wearables with minimal power consumption by the object feature extraction optimization and multi-sensor logic-based attention.

A. IMPACT OF MULTIMODAL OBJECT DETECTION

To quantitatively measure the influence of the object feature extraction, the experimental model exploits the Intersection over Union (IoU) and mean Average Precision (mAP) metrics. The impact of object feature extraction is evaluated by the object detection outcome in this experiment. To implement multimodal object detection, the experimental model transforms a set of RGB images into depth images in the outdoor images of the MS-COCO as well as the Obstacle datasets.

$$\text{IoU} = \frac{|\mathcal{N}_D| \cap |\mathcal{N}_G|}{|\mathcal{N}_D| \cup |\mathcal{N}_G|} \tag{14}$$

IoU is the measure of overlap between the ground truth bounding boxes and predicted bounding boxes, indicating the accuracy of bounding box prediction. In the IoU measurement, $|\mathcal{N}_D|$ and $|\mathcal{N}_G|$ refers to the number of detected

bounding boxes in the image and the number of bounding boxes in the ground truth images.

$$\text{mAP} = \frac{\sum \text{AP of all the classes}}{\text{Total Number of Classes}} \tag{15}$$

mAP is the computation of average precision values at different IoU thresholds for each object category. In addition to the IoU and mAP metrics, the experimental model measures the recall and accuracy metrics to assess the model's performance in detecting all the relevant objects.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{16}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{17}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN} \tag{18}$$

To build the confusion matrix for the object detection task, the experimental model formulates the classification problem in the bounding box prediction. In contrast to the traditional confusion matrix, the object detection confusion matrix is created by matching predicted bounding boxes with ground truth boxes. From the IoU score, the threshold value of the bounding box is assigned as 0.5 to determine the true positives that have higher IoU than the threshold value.

False Negative (FN) refers to the predicted bounding boxes below the IoU threshold. On the other hand, a False positive (FP) indicates that the predicted box is not mapped with any ground truth objects, referring to the fact that the IoU value does not meet the threshold value for all the ground truth boxes. True Negative (TN) is not measured for the object detection task.

Hence, the accuracy metric is the secondary metric for the complete evaluation. As a result, in object or obstacle detection evaluation, True positives refer to the number of correctly detected objects or obstacles, False positives define the number of incorrect obstacle detection, and False negative refers to the number of missed obstacles during the detection.

1) EVALUATION OF OBJECT DETECTION PERFORMANCE

Table 3 evaluates the proposed MuSK in assessing the multimodal object detection performance with the baseline YOLO models and the existing method [36]. The results mentioned in Table 3 are evaluated from the MS-COCO test dataset for object detection.

The object detection performance in the proposed system validates the improved YOLOv7 model-based object feature extraction. The MuSK is assessed using an improved YOLOv7 model in the encoder-decoder image captioning architecture. Compared to the SSD model, the YOLO variants outperform the object detection performance for the MS-COCO dataset under the metrics of mAP, recall, and accuracy in addition to the measure of IoU.

The average measure of IoU value in each image across the detected and filtered objects indicates the accurate localization of the object patches. As presented in Table 3, the proposed approach accomplishes 93.72% mAP and 90.04% accuracy, comparatively higher than the baseline models. This is because the proposed approach examines situational awareness from both RGB and depth images, which aids in accurately recognizing the objects in each image.

TABLE 3. Comparative performance of the object detection tested on the MS-COCO dataset.

Methods and Works	mAP (%)	Recall (%)	Accuracy (%)	IoU (%)
SSD	79.58	78.25	79.75	61.87
YOLOv4	80.18	79.51	81.76	68.93
YOLOv5	82.76	81.53	82.51	79.99
YOLOv7	84.54	83.26	84.03	80.56
YOLOv7 + MobileNetV3	85.92	84.64	83.11	81.44
Guravaiah, K et al. (2023)[36]	88.65	88.96	89.44	83.64
Proposed MuSK	93.72	89.12	90.04	83.86

2) EVALUATION OF OBSTACLE DETECTION PERFORMANCE

In order to assess the performance of obstacle recognition in the outdoor images, the experimental model evaluates the proposed and several existing algorithms on the obstacle dataset, as mentioned in Table 4.

To assess the obstacle detection performance in the encoder-decoder image captioning architecture, the experimental model captures the feature maps of personalized attention score-associated image frames using the output layer, measured for evaluation. The overall accuracy of the proposed MuSK is 90.04%, with an mAP of 93.72% and a recall of 89.12%. Compared to the existing works [50], [51], [52], [53], the significance of the YOLO model associated with transfer learning and layered attention architecture of multimodal cues-based obstacle knowledge learning in the decoder of the proposed system proves its capability for obstacle detection performance. In particular, the IoU metric assesses the accurate localization of the obstacle-bounding boxes about the ground truth bounding box.

Hence, the proposed approach accomplished 83.81% IoU, which is comparatively higher than the existing work [45], even though the generalized MS-COCO features are fine-tuned for the target dataset by the transfer learning due to the obstacle-aware personalized attention score modeling in the encoder and cross-attention modeling in the decoder

TABLE 4. Comparative performance of the obstacle detection tested on the obstacle dataset.

Works	mAP (%)	Recall (%)	Accuracy (%)	IoU (%)
Montalbo F, (2020) [50]	93.14	88.52	88.65	72.51
Yadav S et al., (2020) [51]	85.19	83.88	81.26	79.98
Rachburee N & Punlumjeak W (2021) [52]	86.12	86.54	87.09	80.01
Tang W et al., (2023) [53]	88.54	88.12	85.09	81.08
Proposed MuSK	93.72	89.12	90.04	3.86

of the proposed image captioning system leverages the improved performance in the obstacle dataset.

The comparative works [51], [52] focused on obstacle detection; however, the lack of an enhancement in the traditional YOLO model with the filtered obstacle category for the target dataset leads to an increase in the false positives and false negatives, that is, recognizing the general objects as The obstacles and missing obstacles in the image frames. In addition, the work in [53] validates that the obstacle dataset fails to focus on situational knowledge for accurate obstacle detection. As a result, the situational knowledge-aware logic gate-based personalized attention score modeling enforces the accurate detection of obstacles from the general objects, evaluating the middle of the encoder-decoder architecture for the experimental result illustration. Thus, the proposed approach yields 8.53%, 7.6%, and 5.18% higher mAP than the obstacle detection research [51], [52], and [53], respectively.

B. IMPACT OF IMAGE CAPTIONING MODEL

To evaluate the obstacle alert model, the experimental model employs several metrics, including the Bilingual Evaluation Understudy (BLEU) [54], the Metric for Evaluation of Translation with Explicit Ordering (METEOR) [55], and Consensus-based Image Description Evaluation (CIDEr) [56]. BLEU measures the n-gram co-occurrences between the candidate sentence and reference sentence. METEOR measures the machine translation performance and provides a more accurate evaluation than the BLEU. CIDEr measures the consistency of the image captions from the investigation of each n-tuple based on the Term Frequency Inverse Document Frequency (TF-IDF).

The experimental model investigates the MS-COCO dataset and obstacle dataset for validating the captioning performance, that is, decoder architecture for the visual inputs with the aid of associated ground truth captions in MS-COCO and synthetically tagged captions in the Obstacle dataset. In order to assess the captioning outcome, several existing image captioning researches are compared in Table 5 and Table 6 for BLEU, METEOR, and CIDEr metrics.

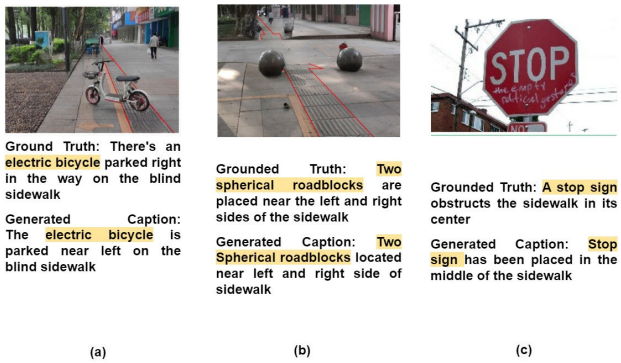

FIGURE 8. (a-c): Sample Input-Output illustration of the proposed obstacle alert system.

Figure 8(a-c) shows the sample images or scenery captured near the blind sidewalk with the generated and ground truth captions.

1) GENERALIZED DATASET - CAPTIONING PERFORMANCE

Results mentioned in Table 5 prove that the proposed approach outperforms several existing comparative research works in the context of image captioning while testing on the generalized MS-COCO dataset for the assessment of the attention significance of the decoder performance.

The comparative image captioning research works [57], [58], [59], [60], [61], [62] generate captions for the images by designing various attention models. However, accomplishing higher performance in all the text assessment metrics of BLEU, METEOR, and CIDEr is more challenging than the proposed approach due to the lack of investigation of several surrounding information about the objects in each image.

The image captioning performance is provided for the evaluation scenarios conducted by the corresponding research works while evaluating the MS-COCO dataset. For the proposed system, the MS-COCO dataset is evaluated under the settings of work [23] to assess the encoder-decoder design of the proposed captioning comparatively.

TABLE 5. Comparative performance of the image captioning tested on the MS-COCO dataset.

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
Xu K et al., (2016) [57]	0.718	0.504	0.357	0.250	0.2304	-
Lu J et al., (2017) [58]	0.742	0.580	0.439	0.332	0.266	1.085
Anderson P et al., (2018) [59]	0.772	-	-	0.362	0.270	1.135
Wang E et al., (2019) [60]	-	-	0.526	0.352	0.232	1.184
Chu Y et al., (2020) [61]	0.731	0.562	0.410	0.326	0.261	0.872
Iwamura K et al., (2021) [23]	0.759	0.599	0.460	0.352	0.267	1.099
Al-Malla M. A et al., (2022) [62]	0.492	0.296	0.174	0.101	0.163	0.390
Proposed MuSK	0.775	0.614	0.488	0.397	0.285	1.197

However, the improved YOLOv7 model as the feature extractor in the encoder-decoder architecture precisely recognizes the object feature maps in the outdoor images rather than determining the overlapping object feature maps.

Hence, the proposed MuSK obtains 0.775 for BLEU-1 and 0.397 for BLEU-4, whereas the comparative work [23] achieves only 0.759 for BLEU-1 and 0.352 for BLEU-5. It proves the ability of the proposed approach to generate highly relevant or similar captions to the ground truth. It is accomplished by the combined situational knowledge in layered attention architecture with the additional focus on the multimodal cues-based obstacle-aware weight matrix in the personalized attention score modeling.

Moreover, the proposed approach yields a comparatively higher METEOR score of 0.285 than the existing image captioning methods, which measures the comprehensive generation of captions for the detected objects through the lexical relationship analysis among the words in the captions.

2) CONTEXTUAL DATASET - CAPTIONING PERFORMANCE

Table 6 compares the image captioning research works in the context of the obstacle alert system for validating the significance of the proposed captions in assisting visually impaired navigation. The object features significantly improve the caption generation with the assistance of the sequential model, such as the LSTM in [23] and [61] and GRU in [62]. In the proposed system, the GRU decoder is implemented as the caption generator with the influence of the customized attention model to highlight obstacles based on situational knowledge.

The experimental outcome illustrates that the proposed image captioning model has outperformed other research works, such as the BLEU, METEOR, and CIDEr. The proposed approach obtains a METEOR score of 0.294 and CIDEr of 1.215 while testing on the obstacle dataset exhibits a significant increase of the comparative image captioning work in [62], respectively. Moreover, the results of the proposed captioning are influenced by the improved YOLOv7 model with transfer learning, providing the object features in terms of the multimodal cues-based personalized attention scores to the decoder model.

The BLEU, METEOR, and CIDEr scores of the proposed obstacle dataset captioning are slightly increased from the generalized caption generation on the MS-COCO dataset mentioned in Table 6, accomplished by the finetuning on the target dataset with the knowledge of the YOLOv7 model through transfer learning. In contrast, the performance of the existing research works [61, 23, 62] has degraded from Table 5 to Table 6 because the attention layer in the existing works is not well designed for obstacle detection. In addition, inaccurate object feature extraction misleads the irrelevant caption generation for the obstacles while testing on the obstacle dataset. Thus, the obstacle-aware personalized attention in the proposed image captioning model rapidly increases the BLEU and METEOR metrics without compromising the CIDEr score.

TABLE 6. Comparative performance of the obstacle captioning tested on the obstacle dataset.

Method	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR	CIDEr
Chu Y et al., (2020) [61]	0.631	0.512	0.354	0.300	0.202	0.801
Iwamura K et al., (2021) [23]	0.642	0.470	0.347	0.313	0.228	1.012
Al-Malla M. A et al., (2022) [62]	0.431	0.243	0.147	0.097	0.125	0.279
Proposed MuSK	0.796	0.678	0.501	0.479	0.294	1.215

In particular, the lack of contextual attention design in the existing research misguides the decoder model during caption generation for the obstacle dataset. Thus, the proposed approach enhances the learning ability of the decoder model through obstacle-aware personalized attention modeling in the encoder and cross-attention modeling in the decoder based on situational knowledge.

VI. CONCLUSION

This work presented the obstacle alert system to the visually impaired, addressing the challenges of single-modality obstacle detection in outdoor environments through multimodal visual and non-visual cues. The proposed approach ensured a lightweight design for the smart wearables by an improved YOLOv7 for object feature extraction and personalized attention design for obstacle alert generation. A layered attention mechanism enabled the contextual fusion of multi-sensory data with visual feature maps by the obstacle-specific knowledge learning from the multi-sensory situational knowledge data. The proposed system's personalized attention score is based on a custom logic gate with dynamically learned obstacle weights from situational knowledge, refining detection accuracy by prioritizing the relevant cues. Finally, cross-attention modeling has leveraged the generation of obstacle-aware captions, enhancing situational awareness. Experimental results have shown that MuSK outperformed the comparative navigation assistance models by yielding 0.294 METEOR score and 1.215 CIDEr score.

Despite the DL-enabled advancements, several limitations hamper the progress of the assistive systems. The assistive alert systems still have a number of issues that restrict their usefulness in practical settings. The current solutions often rely on vision-based models, which are rendered inoperable in messy surroundings, low light levels, and occlusions. Also, real-time cross-modal alignment is a challenge for existing multimodal systems, which results in delays and inconsistent or inaccurate decision-making. Finally, the generalization of these methods is further challenged by wearable device hardware restrictions such as energy consumption and computing overhead.

In future work, a sophisticated, DL-based strategy that combines hierarchical cross-modal learning and real-time cognitive feedback is required to overcome these obstacles and provide VIPs with intelligent, customized, and context-aware assistance. Bridging the gap between DL-driven assistance and human sensory experience is essential for context-aware multimodal assistance systems.

Recent advancements in large vision-language models are more capable of understanding the environment and assisting the VIPs by describing the scene as the alert text. Despite its capabilities, the usefulness of these models is currently understudied to enhance the support of VIPs.

REFERENCES

[1] World Health Organization. (2023). *WHO Releases New Global Estimates on Visual Impairment*. World Health Organization-Regional Office for the Eastern Mediterranean. Accessed: Nov. 28, 2024. [Online]. Available: https://www.emro.who.int/control-and-preventions-of-blindness-and-deafness/announcements/global-estimates-on-visual-impairment.html

[2] G. Cheng, X. Yuan, X. Yao, K. Yan, Q. Zeng, X. Xie, and J. Han, "Towards large-scale small object detection: Survey and benchmarks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13467–13488, Nov. 2023.

[3] B. Kuriakose, R. Shrestha, and F. E. Sandnes, "Tools and technologies for blind and visually impaired navigation support: A review," *IETE Tech. Rev.*, vol. 39, no. 1, pp. 3–18, Jan. 2022.

[4] K. Patel and B. Parmar, "Assistive device using computer vision and image processing for visually impaired; review and current status," *Disab. Rehabil., Assistive Technol.*, vol. 17, no. 3, pp. 290–297, Apr. 2022.

[5] M. Stefanini, M. Cornia, L. Baraldi, S. Cascianelli, G. Fiameni, and R. Cucchiara, "From show to tell: A survey on deep learning-based image captioning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 539–559, Jan. 2023.

[6] R. Kaur and S. Singh, "A comprehensive review of object detection with deep learning," *Digit. Signal Process.*, vol. 132, Jan. 2023, Art. no. 103812.

[7] H. Walle, C. De Runz, B. Serres, and G. Venturini, "A survey on recent advances in AI and vision-based methods for helping and guiding visually impaired people," *Appl. Sci.*, vol. 12, no. 5, p. 2308, Feb. 2022.

[8] Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proc. IEEE Inst. Electr. Electron. Eng.*, vol. 111, no. 3, pp. 257–276, Mar. 2023.

[9] T. Diwan, G. Anirudh, and J. V. Tembhurne, "Object detection using YOLO: Challenges, architectural successors, datasets and applications," *Multimedia Tools Appl.*, vol. 82, no. 6, pp. 9243–9275, Mar. 2023.

[10] P. Xu, G. A. Kennedy, F.-Y. Zhao, W.-J. Zhang, and R. Van Schyndel, "Wearable obstacle avoidance electronic travel aids for blind and visually impaired individuals: A systematic review," *IEEE Access*, vol. 11, pp. 66587–66613, 2023.

[11] B. Kuriakose, R. Shrestha, and F. E. Sandnes, "Multimodal navigation systems for users with visual impairments—A review and analysis," *Multimodal Technol. Interact.*, vol. 4, no. 4, p. 73, Oct. 2020.

[12] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "DINO: DETR with improved DeNoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.

[13] K. Wang, Y. Wang, S. Zhang, Y. Tian, and D. Li, "SLMS-SSD: Improving the balance of semantic and spatial information in object detection," *Expert Syst. Appl.*, vol. 206, Nov. 2022, Art. no. 117682.

[14] T. Sharma, B. Debaque, N. Duclos, A. Chehri, B. Kinder, and P. Fortier, "Deep learning-based object detection and scene perception under bad weather conditions," *Electronics*, vol. 11, no. 4, p. 563, Feb. 2022.

[15] R. Duan, H. Deng, M. Tian, Y. Deng, and J. Lin, "SODA: A large-scale open site object detection dataset for deep learning in construction," *Autom. Construct.*, vol. 142, Oct. 2022, Art. no. 104499.

[16] M. E. Salman, G. Ç. Çakar, J. Azimjonov, M. Kösem, and İ. H. Cedimoğlu, "Automated prostate cancer grading and diagnosis system using deep learning-based YOLO object detection algorithm," *Expert Syst. Appl.*, vol. 201, Sep. 2022, Art. no. 117148.

[17] Y. Wang, B. Xiao, A. Bouferguene, M. Al-Hussein, and H. Li, "Vision-based method for semantic information extraction in construction by integrating deep learning object detection and image captioning," *Adv. Eng. Informat.*, vol. 53, Aug. 2022, Art. no. 101699.

[18] C.-W. Kuo and Z. Kira, "Beyond a pre-trained object detector: Cross-modal textual and visual context for image captioning," 2022, *arXiv:2205.04363*.

[19] W. Liu, G. Ren, R. Yu, S. Guo, J. Zhu, and L. Zhang, "Image-adaptive YOLO for object detection in adverse weather conditions," 2021, *arXiv:2112.08088*.

[20] J. Lee and K.-I. Hwang, "YOLO with adaptive frame control for real-time object detection applications," *Multimedia Tools Appl.*, vol. 81, no. 25, pp. 36375–36396, Oct. 2022.

[21] Y. Lyu, Z. Liu, H. Li, D. Guo, and Y. Fu, "A real-time and lightweight method for tiny airborne object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 3016–3025.

[22] J. Bai, Z. Liu, Y. Lin, Y. Li, S. Lian, and D. Liu, "Wearable travel aid for environment perception and navigation of visually impaired people," *Electronics*, vol. 8, no. 6, p. 697, Jun. 2019.

[23] K. Iwamura, J. Y. Louhi Kasahara, A. Moro, A. Yamashita, and H. Asama, "Image captioning using motion-CNN with object detection," *Sensors*, vol. 21, no. 4, p. 1270, Feb. 2021.

[24] A. Ben Atitallah, Y. Said, M. A. Ben Atitallah, M. Albekairi, K. Kaaniche, and S. Boubaker, "An effective obstacle detection system using deep learning advantages to aid blind and visually impaired navigation," *Ain Shams Eng. J.*, vol. 15, no. 2, Feb. 2024, Art. no. 102387.

[25] L. Bougheloum, M. B. Salah, and M. Bettayeb, "Outdoor obstacle detection for visually impaired using AI technique," in *Proc. ASU Int. Conf. Emerg. Technol. Sustainability Intell. Syst. (ICETSIS)*, Jan. 2024, pp. 628–633.

[26] T. Saidani, "Deep learning approach: YOLOv5-based custom object detection," *Eng., Technol. Appl. Sci. Res.*, vol. 13, no. 6, pp. 12158–12163, Dec. 2023.

[27] A. K. Mtasher and E. H. J. Al-Wakel, "Custom object detection using transfer learning with pretrained models for improved detection techniques," *J. La Multiapp*, vol. 5, no. 1, pp. 10–18, Feb. 2024.

[28] Z. Bauer, A. Dominguez, E. Cruz, F. Gomez-Donoso, S. Orts-Escolano, and M. Cazorla, "Enhancing perception for the visually impaired with deep learning techniques and low-cost wearable sensors," *Pattern Recognit. Lett.*, vol. 137, pp. 27–36, Sep. 2020.

[29] Z. Xie, Z. Li, Y. Zhang, J. Zhang, F. Liu, and W. Chen, "A multi-sensory guidance system for the visually impaired using YOLO and ORB-SLAM," *Information*, vol. 13, no. 7, p. 343, Jul. 2022.

[30] D. Chaudhary, A. Mathur, A. Chauhan, and A. Gupta, "Assistive object recognition and obstacle detection system for the visually impaired using YOLO," in *Proc. 13th Int. Conf. Cloud Comput., Data Sci. Eng. (Confluence)*, Jan. 2023, pp. 353–358.

[31] S. Shadi, S. Hadi, M. A. Nazari, and W. Hardt, "Outdoor navigation for visually impaired based on deep learning," in *Proc. CEUR Workshop*, vol. 2514, 2019, pp. 97–406.

[32] W.-J. Chang, L.-B. Chen, M.-C. Chen, J.-P. Su, C.-Y. Sie, and C.-H. Yang, "Design and implementation of an intelligent assistive system for visually impaired people for aerial obstacle avoidance and fall detection," *IEEE Sensors J.*, vol. 20, no. 17, pp. 10199–10210, Sep. 2020.

[33] B. Pydala, T. P. Kumar, and K. K. Baseer, "Smart_Eye: A navigation and obstacle detection for visually impaired people through smart app," *J. Appl. Eng. Technol. Sci.*, vol. 4, no. 2, pp. 992–1011, Jun. 2023.

[34] Y. Said, M. Atri, M. A. Albahar, A. Ben Atitallah, and Y. A. Alsariera, "Obstacle detection system for navigation assistance of visually impaired people based on deep learning techniques," *Sensors*, vol. 23, no. 11, p. 5262, Jun. 2023.

[35] S. Suman, S. Mishra, K. S. Sahoo, and A. Nayyar, "Vision navigator: A smart and intelligent obstacle recognition model for visually impaired users," *Mobile Inf. Syst.*, vol. 2022, pp. 1–15, Mar. 2022.

[36] K. Guravaiah, Y. S. Bhavadeesh, P. Shwejan, A. H. Vardhan, and S. Lavanya, "Third eye: Object recognition and speech generation for visually impaired," *Proc. Comput. Sci.*, vol. 218, pp. 1144–1155, Jan. 2023.

[37] X. Xu, Y. Li, G. Wu, and J. Luo, "Multi-modal deep feature learning for RGB-D object detection," *Pattern Recognit.*, vol. 72, pp. 300–313, Dec. 2017.

[38] J. Kim and J. Cho, "Exploring a multimodal mixture-of-YOLOs framework for advanced real-time object detection," *Appl. Sci.*, vol. 10, no. 2, p. 612, Jan. 2020.

[39] Y. Luo, C. Zhang, M. Zhao, H. Zhou, and J. Sun, "Where, what, whether: Multimodal learning meets pedestrian detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 14065–14073.

[40] Y. Bazi, H. Alhichri, N. Alajlan, and F. Melgani, "Scene description for visually impaired people with multi-label convolutional SVM networks," *Appl. Sci.*, vol. 9, no. 23, p. 5062, Nov. 2019.

[41] R. C. Joshi, S. Yadav, M. K. Dutta, and C. M. Travieso-Gonzalez, "Efficient multi-object detection and smart navigation using artificial intelligence for visually impaired people," *Entropy*, vol. 22, no. 9, p. 941, Aug. 2020.

[42] A. Mustafa Hilal, F. Alrowais, F. N. Al-Wesabi, and R. Marzouk, "Red deer optimization with artificial intelligence enabled image captioning system for visually impaired people," *Comput. Syst. Sci. Eng.*, vol. 46, no. 2, pp. 1929–1945, 2023, doi: 10.32604/csse.2023.035529.

[43] F. E. El-Taher, L. Miralles-Pechuán, J. Courtney, K. Millar, C. Smith, and S. Mckeever, "A survey on outdoor navigation applications for people with visual impairments," *IEEE Access*, vol. 11, pp. 14647–14666, 2023.

[44] C.-Y. Wang, A. Bochkovskiy, and H.-Y.-M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7464–7475.

[45] J. Dai, H. Qi, Y. Xiong, Y. Li, G. Zhang, H. Hu, and Y. Wei, "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773, doi: 10.1109/ICCV.2017.89.

[46] Z. Zohourianshahzadi and J. K. Kalita, "Neural attention for image captioning: Review of outstanding methods," *Artif. Intell. Rev.*, vol. 55, no. 5, pp. 3833–3862, Jun. 2022.

[47] A. Meliones, C. Filios, and J. Llorente, "Reliable ultrasonic obstacle recognition for outdoor blind navigation," *Technologies*, vol. 10, no. 3, p. 54, Apr. 2022, doi: 10.3390/technologies10030054.

[48] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, 2014, pp. 740–755, doi: 10.1007/978-3-319-10602-1_48.

[49] (2024). *Obstacle Dataset*. Accessed: Apr. 2024. [Online]. Available: https://github.com/TW0521/Obstacle-Dataset

[50] F. J. P. Montalbo, "A computer-aided diagnosis of brain tumors using a finetuned YOLO-based model with transfer learning," *KSII Trans. Internet Inf. Syst.*, vol. 14, no. 12, pp. 4816–4834, 2020.

[51] S. Yadav, R. C. Joshi, M. K. Dutta, M. Kiac, and P. Sikora, "Fusion of object recognition and obstacle detection approach for assisting visually challenged person," in *Proc. 43rd Int. Conf. Telecommun. Signal Process. (TSP)*, Jul. 2020, pp. 537–540, doi: 10.1109/TSP49548.2020.9163434.

[52] N. Rachburee and W. Punlumjeak, "An assistive model of obstacle detection based on deep learning: YOLOv3 for visually impaired people," *Int. J. Electr. Comput. Eng.*, vol. 11, no. 4, p. 3434, Aug. 2021.

[53] W. Tang, D.-E. Liu, X. Zhao, Z. Chen, and C. Zhao, "A dataset for the recognition of obstacles on blind sidewalk," *Universal Access Inf. Soc.*, vol. 22, no. 1, pp. 69–82, Mar. 2023.

[54] K. Papineni, S. Roukos, T. Ward, and W. J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meet. Assoc. Comput. Linguistics*, 2002, pp. 311–318.

[55] M. Denkowski and A. Lavie, "Meteor universal: Language specific translation evaluation for any target language," in *Proc. 9th Workshop Stat. Mach. Transl.*, 2014, pp. 376–380.

[56] G. O. dos Santos, E. L. Colombini, and S. Avila, "CIDEr-R: Robust consensus-based image description evaluation," in *Proc. 7th Workshop Noisy User-Generated Text (W-NUT)*, 2021, pp. 351–360.

[57] K. Xu, J. Ba, R. Kiros, K. Cho, A. Courville, R. Salakhutdinov, R. Zemel, and Y. Bengio, "Show, attend and tell: Neural image caption generation with visual attention," 2015, *arXiv:1502.03044*.

[58] X. Liu and Q. Xu, "Adaptive attention-based high-level semantic introduction for image caption," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 16, no. 4, pp. 1–22, Nov. 2020.

[59] P. Anderson, X. He, C. Buehler, D. Teney, M. Johnson, S. Gould, and L. Zhang, "Bottom-up and top-down attention for image captioning and visual question answering," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6077–6086, doi: 10.1109/CVPR.2018.00636.

[60] E. K. Wang, X. Zhang, F. Wang, T.-Y. Wu, and C.-M. Chen, "Multi-layer dense attention model for image caption," *IEEE Access*, vol. 7, pp. 66358–66368, 2019.

[61] Y. Chu, X. Yue, L. Yu, M. Sergei, and Z. Wang, "Automatic image captioning based on ResNet50 and LSTM with soft attention," *Wireless Commun. Mobile Comput.*, vol. 2020, pp. 1–7, Oct. 2020.

[62] M. A. Al-Malla, A. Jafar, and N. Ghneim, "Image captioning model using attention and object features to mimic human image understanding," *J. Big Data*, vol. 9, no. 1, p. 20, Dec. 2022.